

I can calculate Ac-225 activity. I need help deciding what makes that output scientifically useful.

I want the production output to be useful without turning it into a clinical claim.

I am a 12th grader at Glen Burnie High School in Maryland, and I built a reduced model of one Ra-226 to Ac-225 production pathway after learning about targeted alpha therapy. Your work in alpha-particle therapy, radiopharmaceutical dosimetry, modeling, and treatment planning is a close match to the question I am trying to answer. My concern is that a mathematically correct activity value can still be misleading if I report it at the wrong time, use the wrong impurity definition, or connect it to treatment in a way the project has not earned.

PERSONAL NOTE FOR

Dr. Robert F. Hobbs

Johns Hopkins Radiation Oncology and
Molecular Radiation Sciences

What the model currently gives me

- Predicted Ac-225 atoms and activity throughout irradiation and cooling.
- Predicted trace Ac-227 from the reduced side pathway.
- A candidate schedule and a stated calculation time.
- Synthetic error statistics and a warning when inputs leave the studied range.

What those numbers still do not tell me

- Which reference time makes a production comparison fair.
- Whether impurity should be reported by activity, atoms, mass, or another convention.
- How daughter products and measurement uncertainty should be stated.
- What information is useful to a medical reader without implying dose, safety, or clinical validation.

THE REPORTING CHOICE I AM TRYING NOT TO GUESS

I found an IAEA example that uses a 2% Ac-227/Ac-225 activity specification, but I am not treating it as a universal medical or regulatory limit. I would rather cite the right convention and clearly label what is only an example. The project has no patient data, dose model, or clinical validation.

What I would value from you

Could you recommend one or two public references that show how an Ac-225 production or product-quality study should report activity, reference time, radionuclidic impurity, daughter equilibrium, and uncertainty? If there is a better person at Hopkins for this exact question, a referral would also help.

How this changes my project

I would use the reference to define the output before ranking schedules, remove any unsupported purity claim, and explain the medical relevance in plain language while keeping the final claim strictly about production-side screening.